Lab 1: Intro to R

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Loading pacakges using library function.

```
library(tidyverse)
library(openintro)
```

Arbuthnot Data set:

```
arbuthnot
```

```
## # A tibble: 82 x 3
##     year  boys girls
##    <int> <int> <int>
##  1  1629  5218  4683
##  2  1630  4858  4457
##  3  1631  4422  4102
##  4  1632  4994  4590
##  5  1633  5158  4839
##  6  1634  5035  4820
##  7  1635  5106  4928
##  8  1636  4917  4605
##  9  1637  4703  4457
## 10  1638  5359  4952
## # i 72 more rows
```

Exercise 1

1. What command would you use to extract just the counts of girls baptized?

- To know the counts of the girls baptized each year we use **$** sign to extract the particular column of the data set in this case the girls column and we can see the count of the girls in each year.
- To know the total count of the girls baptized we use **Sum()** function here along with the **$**.

```
arbuthnot$girls
```

```
##  [1] 4683 4457 4102 4590 4839 4820 4928 4605 4457 4952 4784 5332 5200 4910 4617
## [16] 3997 3919 3395 3536 3181 2746 2722 2840 2908 2959 3179 3349 3382 3289 3013
```

```
## [31] 2781 3247 4107 4803 4881 5681 4858 4319 5322 5560 5829 5719 6061 6120 5822
## [46] 5738 5717 5847 6203 6033 6041 6299 6533 6744 7158 7127 7246 7119 7214 7101
## [61] 7167 7302 7392 7316 7483 6647 6713 7229 7767 7626 7452 7061 7514 7656 7683
## [76] 5738 7779 7417 7687 7623 7380 7288
```

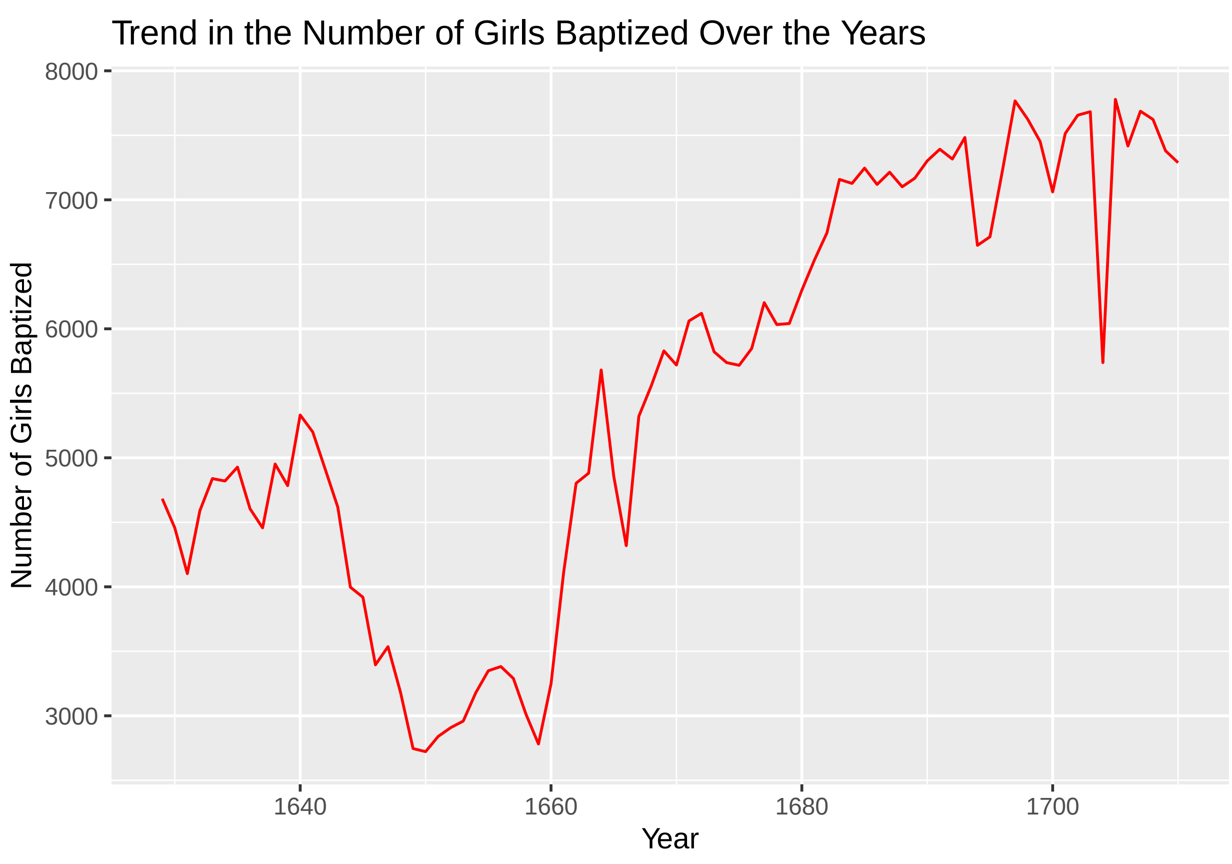
```r
sum(arbuthnot$girls)
```

```
## [1] 453841
```

Exercise 2

2. Is there an apparent trend in the number of girls baptized over the years? How would you describe it?

```r
ggplot(data = arbuthnot, aes(x = year, y = girls)) +
geom_line(color='red')+labs(title = "Trend in the Number of Girls Baptized Over the Years",
       x = "Year",
       y = "Number of Girls Baptized")
```



Overall Description:

There seems to be a *fluctuating trend* with the number of girls that are been baptized between the time span of 1620's to 1720's.

Specific Observations:

- *Early Years (1620-1650):* There is a noticeable decline in the number of girls baptized, reaching a low point around the mid-1650s.
- *Mid-1650s to 1680s:* After the decline, there is a sharp and dramatic increase in the number of girls baptized. This upward trend continues, with some fluctuations in between.
- *Late 1680s to Early 1700s:* The trend becomes more stable with a generally high number of baptisms, though there are a few notable drops, particularly around the 1700s to 1710s.

Exercise 3

Creating the new Variables for the data set using **piping %>% and Mutate()**

```
# Insert code for Exercise 3 here
arbuthnot$boys + arbuthnot$girls
```

```
##  [1]  9901  9315  8524  9584  9997  9855 10034  9522  9160 10311 10150 10850
## [13] 10670 10370  9410  8104  7966  7163  7332  6544  5825  5612  6071  6128
## [25]  6155  6620  7004  7050  6685  6170  5990  6971  8855 10019 10292 11722
## [37]  9972  8997 10938 11633 12335 11997 12510 12563 11895 11851 11775 12399
## [49] 12626 12601 12288 12847 13355 13653 14735 14702 14730 14694 14951 14588
## [61] 14771 15211 15054 14918 15159 13632 13976 14861 15829 16052 15363 14639
## [73] 15616 15687 15448 11851 16145 15369 16066 15862 15220 14928
```

```
# As my Arbuthnot already has the total column i am creating a new column total2
arbuthnot <- arbuthnot %>%
mutate(total2 = boys + girls)
```

Where is the new variable?

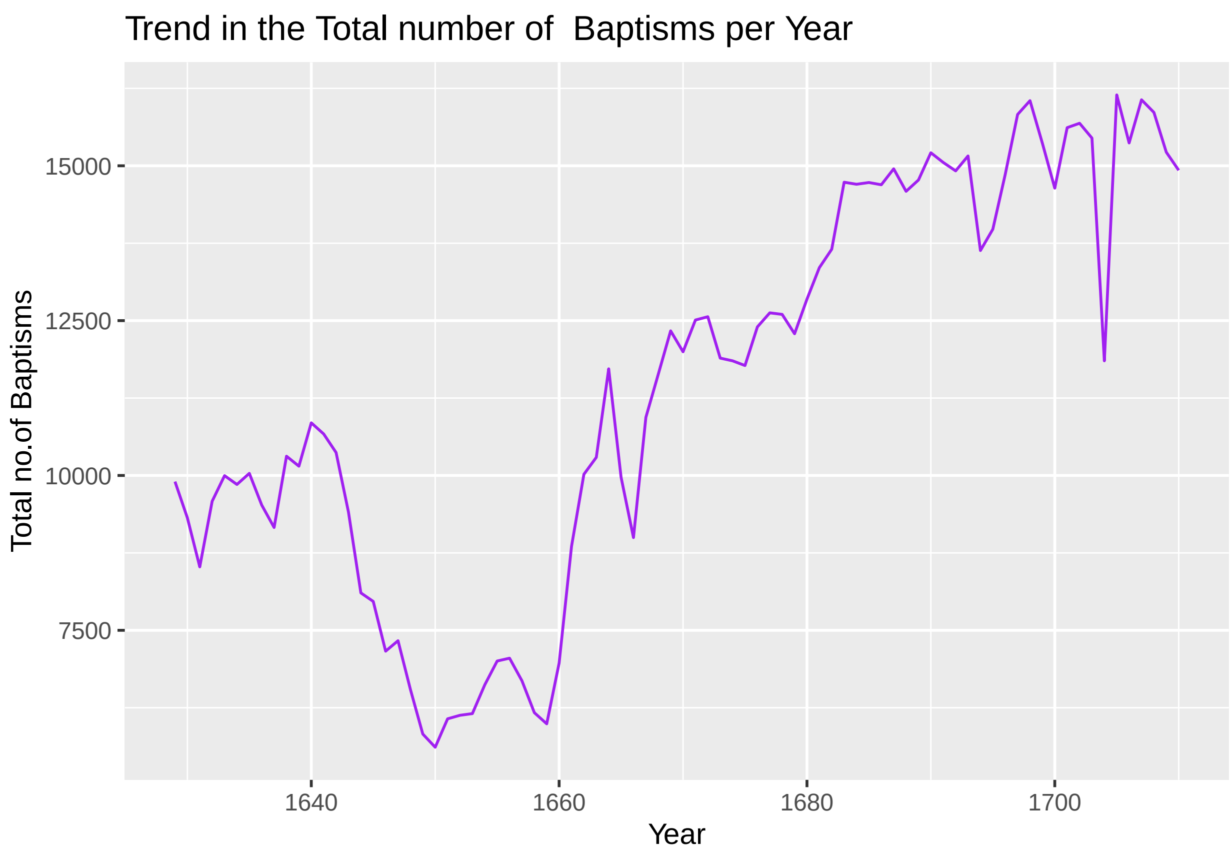
```
names(arbuthnot)
```

```
## [1] "year"   "boys"   "girls"  "total2"
```

Over here we can observe that the new variables are added at the end of the dataset.

Line Plot for Total no. of baptisms per year

```
ggplot(data = arbuthnot, aes(x = year, y = total2)) +
geom_line(color='purple')+labs(title = "Trend in the Total number of  Baptisms per Year",
       x = "Year",
       y = "Total no.of Baptisms")
```



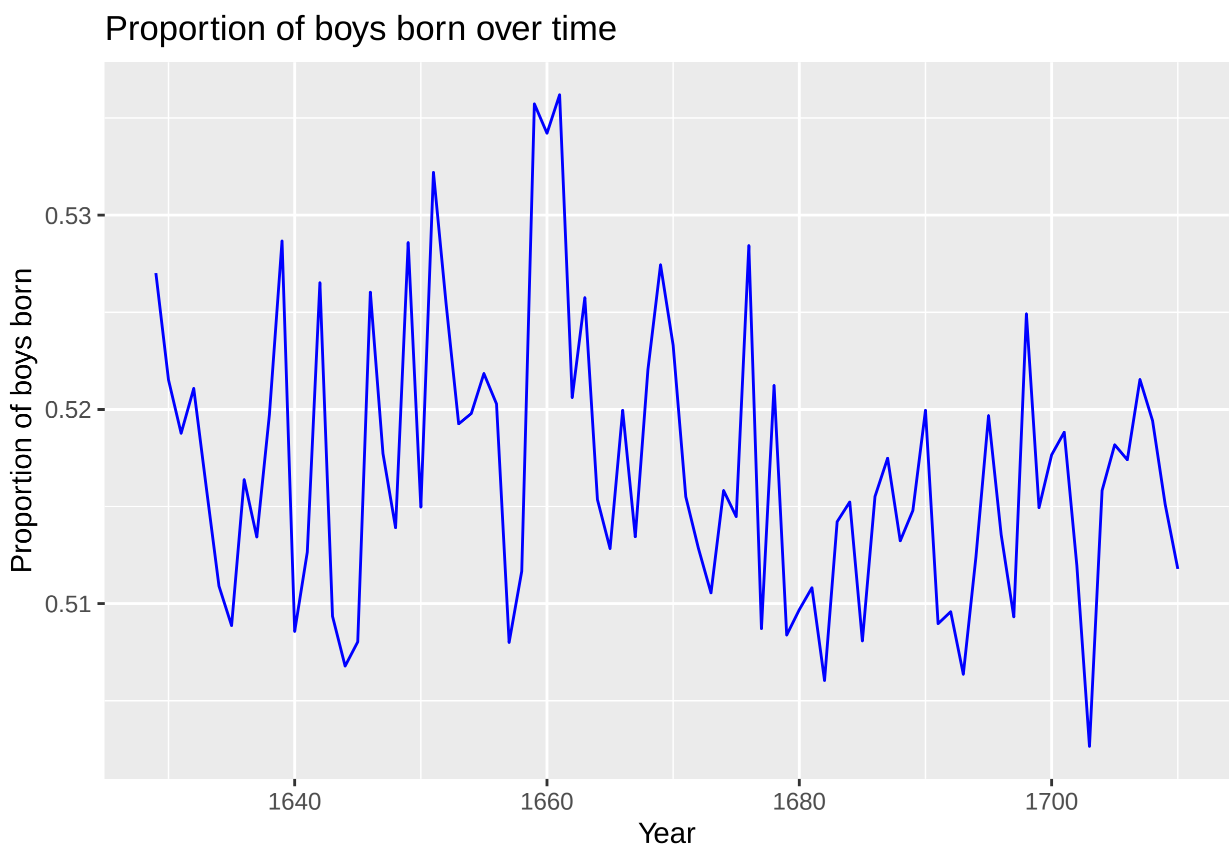
Over here we can observe that it follows the similar trend of the girls baptism. **It has a fluctuating trend over years.**

```r
arbuthnot <- arbuthnot %>%
mutate(boy_to_girl_ratio_2 = boys / girls)
```

```r
arbuthnot <- arbuthnot %>%
mutate(boy_ratio_2 = boys / total2)
```

```r
arbuthnot <- arbuthnot %>%
mutate(girl_ratio = girls / total2)
```

```r
ggplot(data = arbuthnot, aes(x = year, y = boy_ratio_2)) +geom_line(color='blue')+labs(title = "Proporti
```



Observations

- *Fluctuating Proportion:* The proportion of boys born each year fluctuates quite significantly over the time period from 1629 to 1720.
- *Proportion Around 0.51 to 0.53:* The proportion mostly oscillates between approximately 0.51 and 0.53.
- Even though we observe there are higher no of boys baptized in the year 1655's to 1660's we also observed in the previous graphs that there was a dip in the trend for no of girls baptized and total no of children baptised.

Exercise 4

Loading Present Data Set:

```
present
```

```
## # A tibble: 63 x 3
##      year    boys   girls
##     <dbl>   <dbl>   <dbl>
##   1  1940 1211684 1148715
##   2  1941 1289734 1223693
```

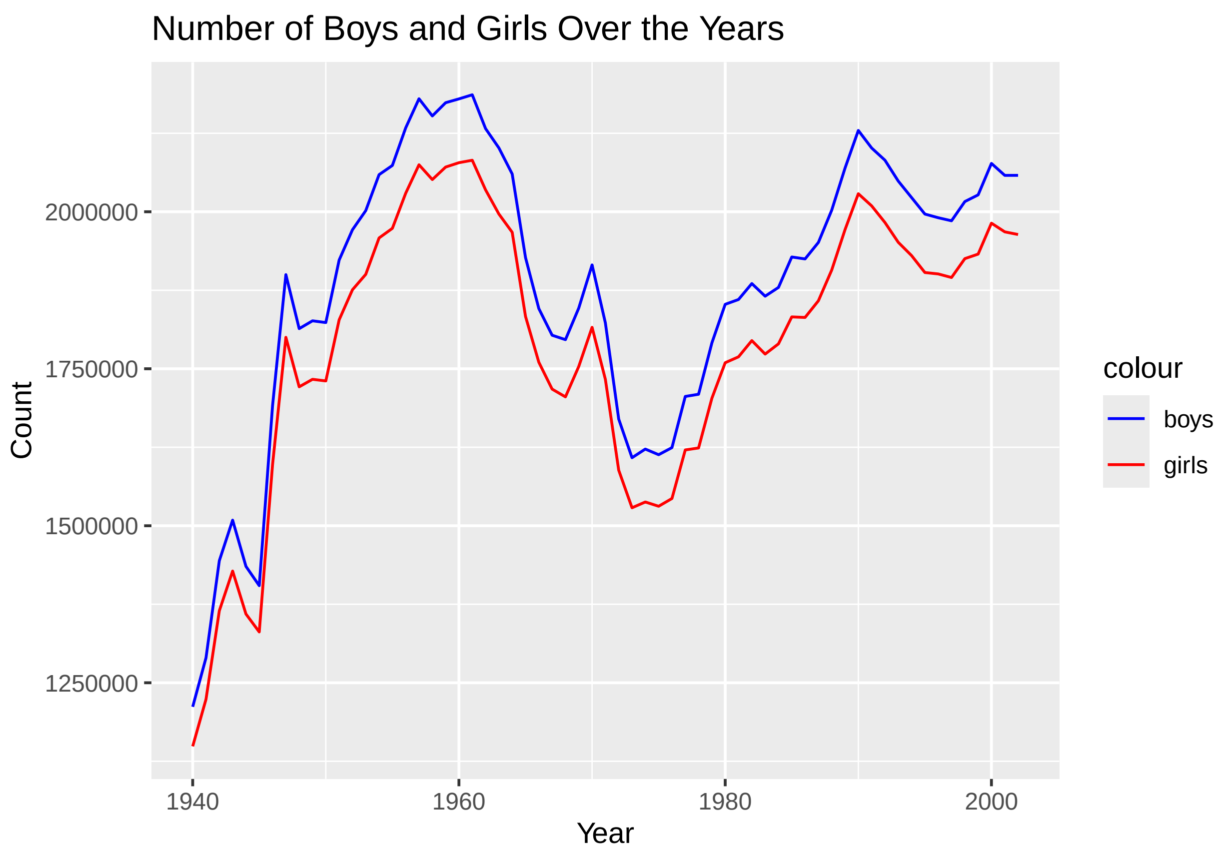
```
## 3  1942 1444365 1364631
## 4  1943 1508959 1427901
## 5  1944 1435301 1359499
## 6  1945 1404587 1330869
## 7  1946 1691220 1597452
## 8  1947 1899876 1800064
## 9  1948 1813852 1721216
## 10  1949 1826352 1733177
## # i 53 more rows
```

```
present %>%
summarize(min = min(boys),
max = max(boys)
)
```

```
## # A tibble: 1 x 2
##       min     max
##     <dbl>   <dbl>
## 1 1211684 2186274
```

What years are included in this data set? What are the dimensions of the data frame? What are the variable (column) names?

```
# Plotting boys and girls on the same plot with different colors
ggplot(data = present, aes(x = year)) +
  geom_line(aes(y = boys, color = "boys")) +
  geom_line(aes(y = girls, color = "girls")) +
  scale_color_manual(values = c("boys" = "blue", "girls" = "red")) +
  labs(title = "Number of Boys and Girls Over the Years",
       x = "Year",
       y = "Count",
       legend="color")
```



1. In the data set the years from 1940 - early 2000's were included.

```
present
```

```
## # A tibble: 63 x 3
##     year    boys   girls
##    <dbl>   <dbl>   <dbl>
##  1  1940 1211684 1148715
##  2  1941 1289734 1223693
##  3  1942 1444365 1364631
##  4  1943 1508959 1427901
##  5  1944 1435301 1359499
##  6  1945 1404587 1330869
##  7  1946 1691220 1597452
##  8  1947 1899876 1800064
##  9  1948 1813852 1721216
## 10  1949 1826352 1733177
## # i 53 more rows
```

2. The data frame has **63 rows and 3 columns** namely year,boys and girls.
3. Here we have three variable columns **year, boys and girls**

Exercise 5

How do these counts compare to Arbuthnot's? Are they of a similar magnitude?

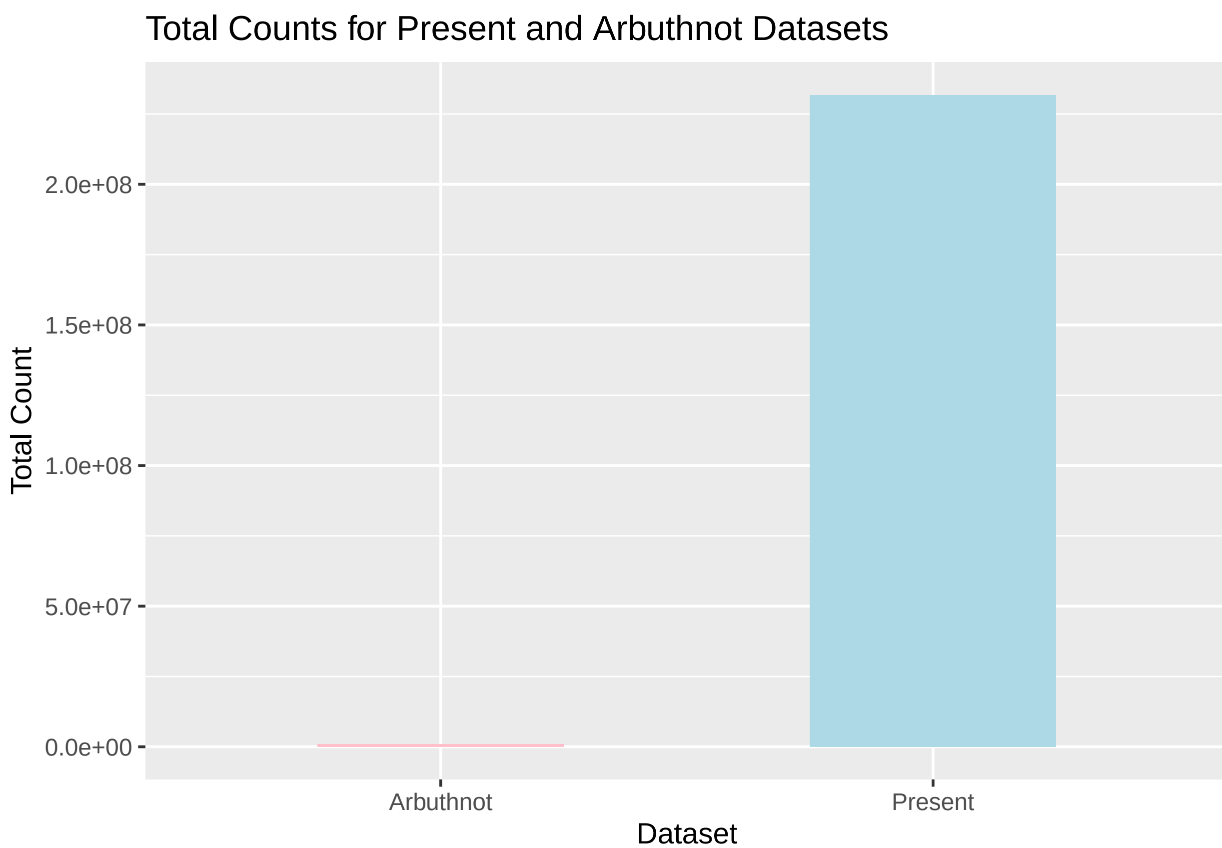
```r
# Insert code for Exercise 5 here
#present$boys + present$girls
present <- present %>%
mutate(total = boys + girls)
present %>% summarize(min = min(total),max = max(total))
```

```
## # A tibble: 1 x 2
##       min     max
##     <dbl>   <dbl>
## 1 2360399 4268326
```

```r
arbuthnot %>% summarize(min=min(total2),max=max(total2))
```

```
## # A tibble: 1 x 2
##     min   max
##   <int> <int>
## 1  5612 16145
```

```r
# Calculate total sums for present and arbuthnot datasets
present_total <- present %>% summarize(total_sum = sum(total))
arbuthnot_total <- arbuthnot %>% summarize(total_sum = sum(total2))

# Combine the results into a format suitable for ggplot2
ggplot() +
  geom_bar(aes(x = "Present", y = present_total$total_sum), stat = "identity", fill = "lightblue", widt
  geom_bar(aes(x = "Arbuthnot", y = arbuthnot_total$total_sum), stat = "identity", fill = "pink", width
  labs(title = "Total Counts for Present and Arbuthnot Datasets",
       x = "Dataset",
       y = "Total Count")
```

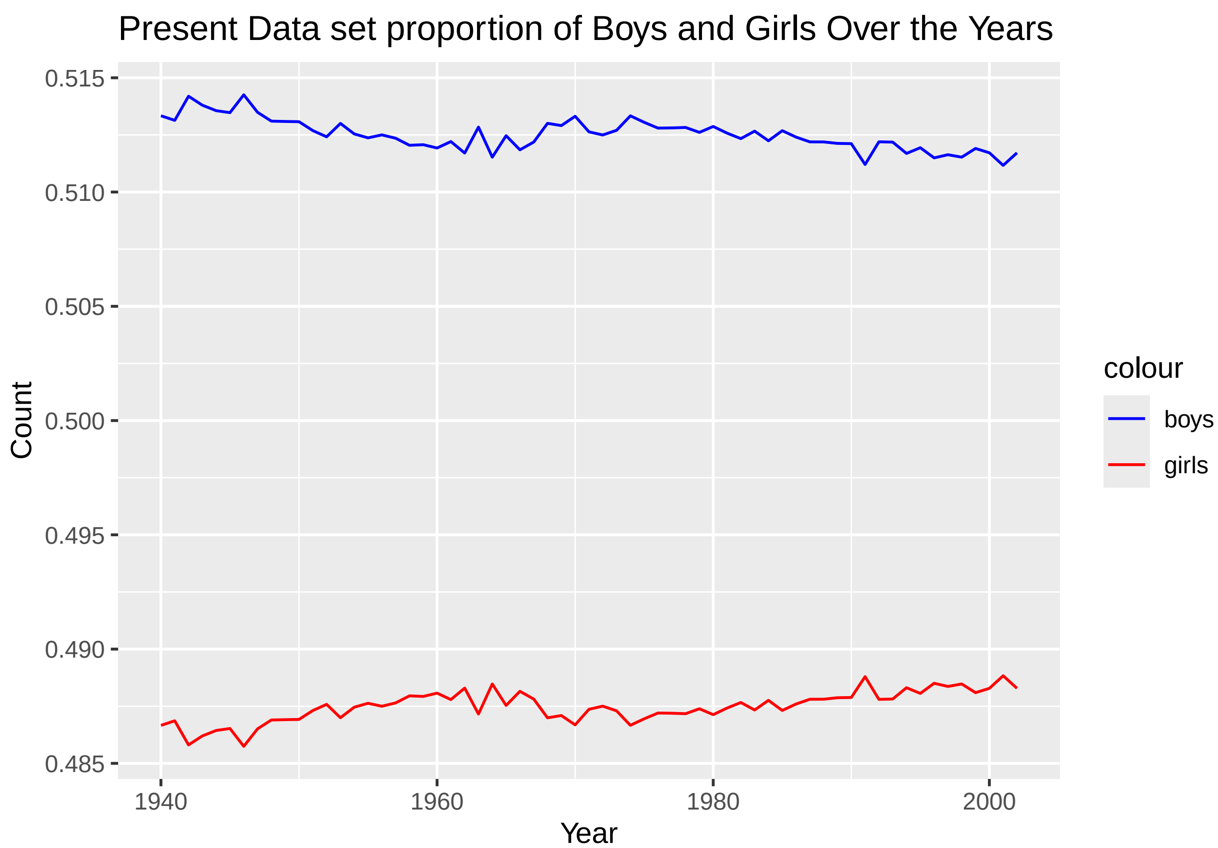


Based on the above graph we can say that the magnitude of the present is way more higher than that of the arbuthnot.

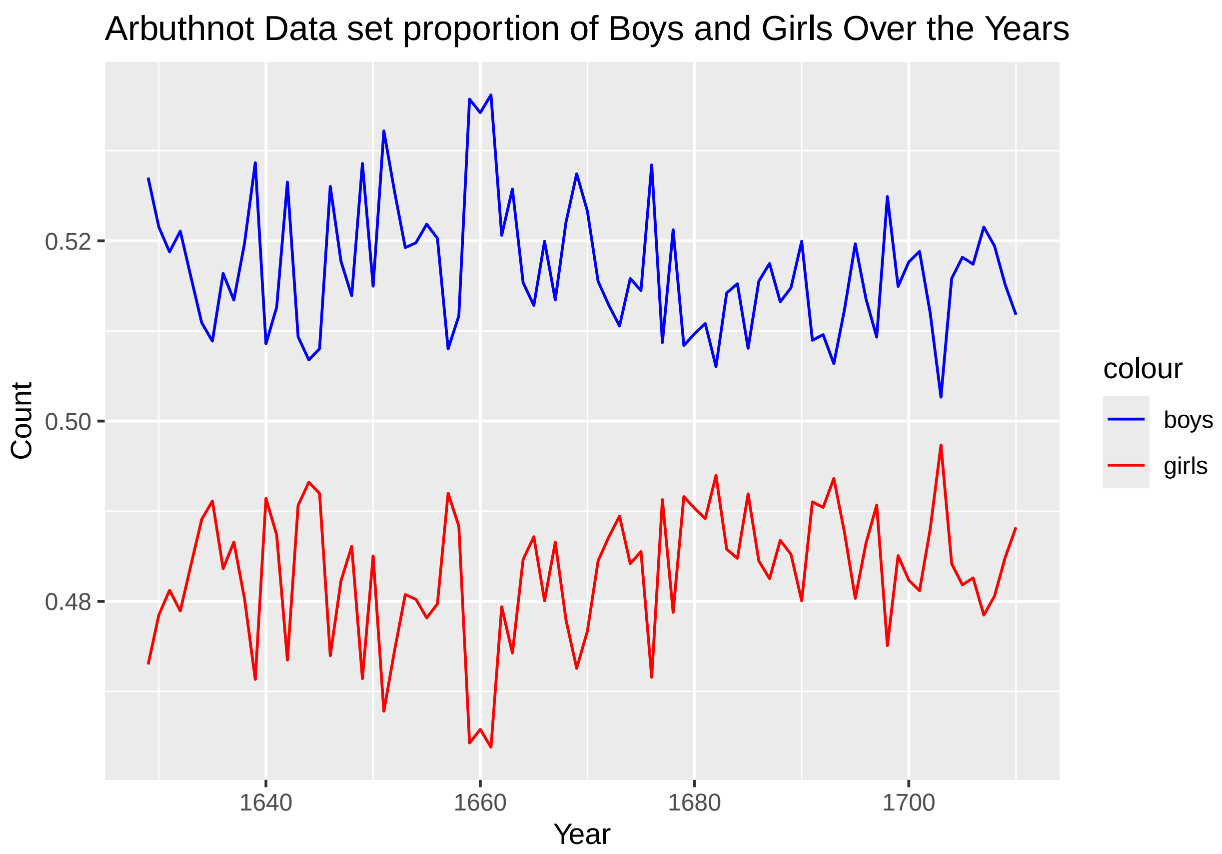
Exercise 6

Make a plot that displays the proportion of boys born over time. What do you see? Does Arbuthnot's observation about boys being born in greater proportion than girls hold up in the U.S.? Include the plot in your response. Hint: You should be able to reuse your code from Exercise 3 above, just replace the name of the data frame.

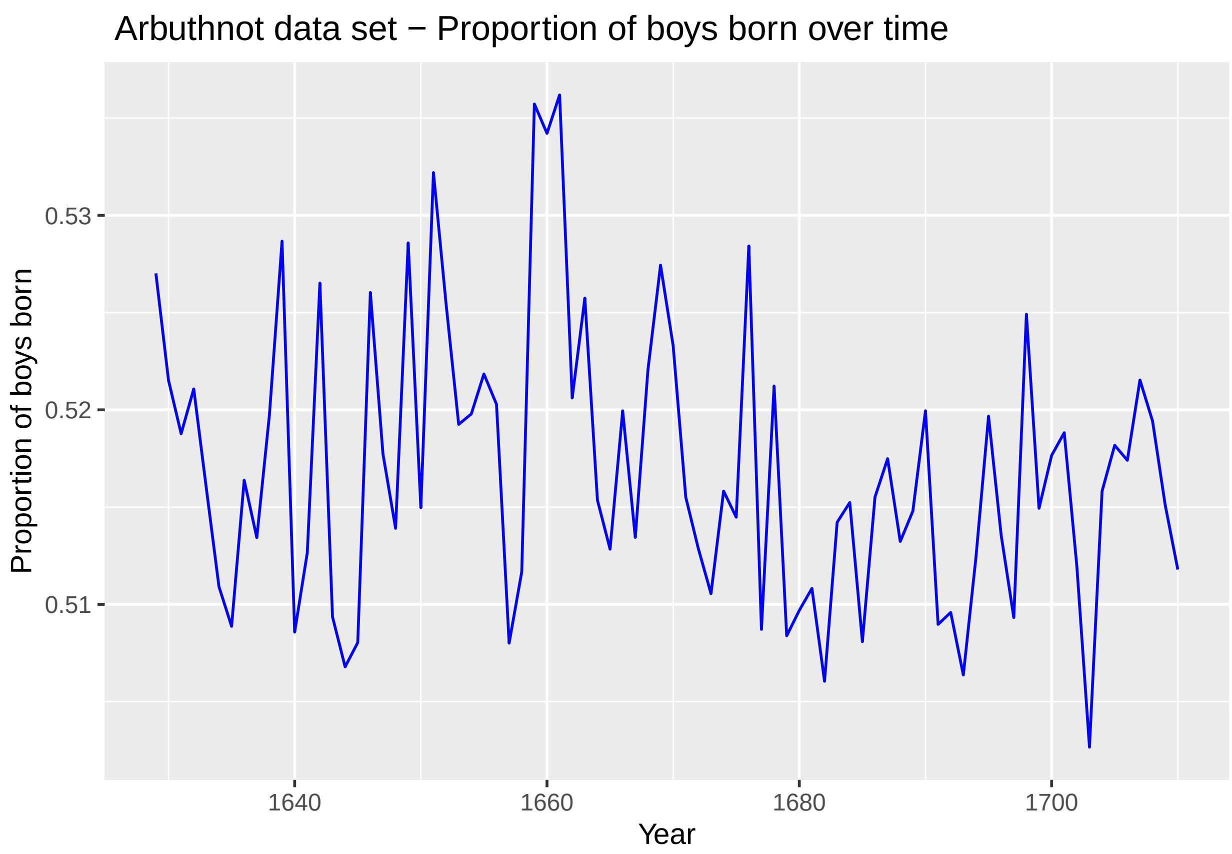
```
# Insert code for Exercise 6 here
present <- present %>%
mutate(present_boy_ratio = boys/total)
present <- present %>%
mutate(present_girl_ratio=girls/total)

ggplot(data = present, aes(x = year)) +
  geom_line(aes(y = present_boy_ratio, color = "boys")) +
  geom_line(aes(y = present_girl_ratio, color = "girls")) +
  scale_color_manual(values = c("boys" = "blue", "girls" = "red")) +
  labs(title = "Present Data set proportion of Boys and Girls Over the Years",
       x = "Year",
       y = "Count",
       legend="color")
```



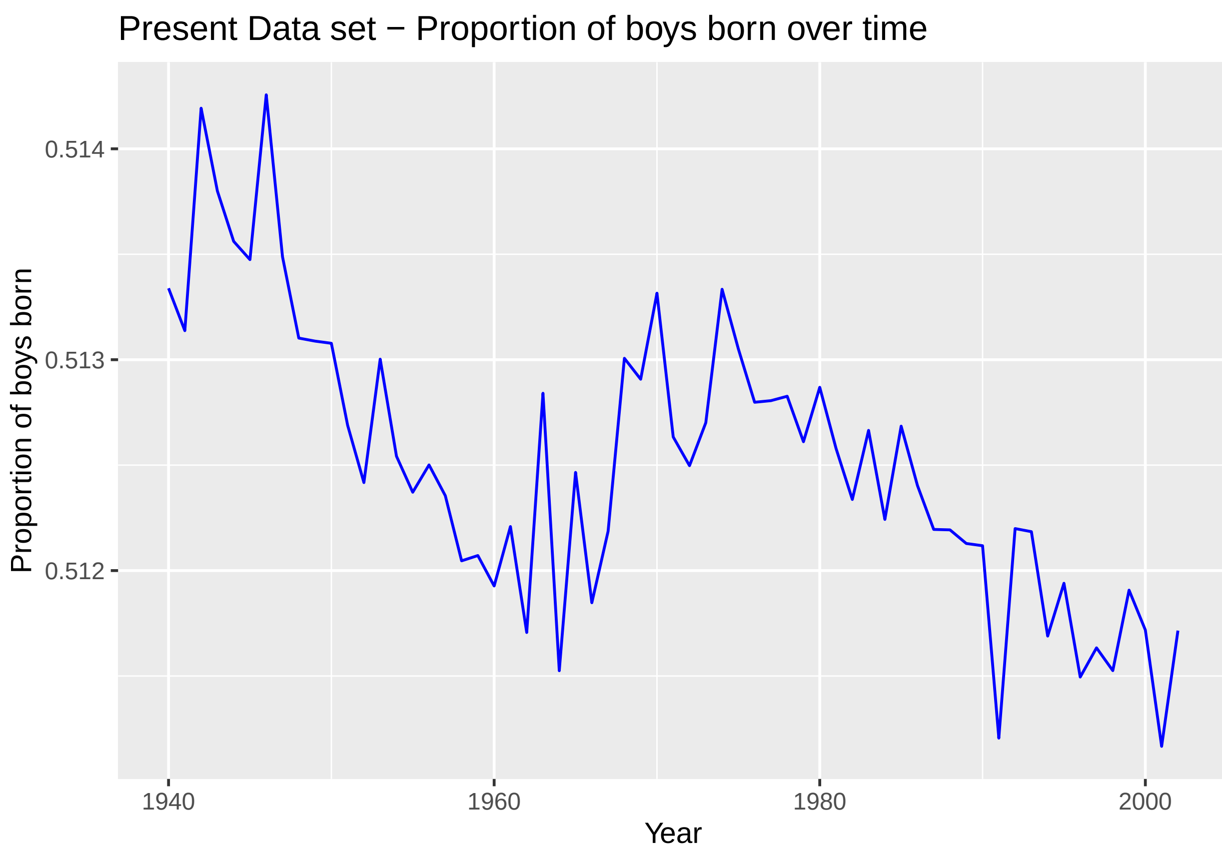

```r
ggplot(data = arbuthnot, aes(x = year)) +
  geom_line(aes(y = boy_ratio_2, color = "boys")) +
  geom_line(aes(y = girl_ratio, color = "girls")) +
  scale_color_manual(values = c("boys" = "blue", "girls" = "red")) +
  labs(title = "Arbuthnot Data set proportion of Boys and Girls Over the Years",
       x = "Year",
       y = "Count",
       legend="color")
```




```
ggplot(data = arbuthnot, aes(x = year, y = boy_ratio_2)) +geom_line(color='blue')+labs(title = " Arbuthn
```




```
ggplot(data = present, aes(x = year, y = present_boy_ratio)) +geom_line(color='blue')+labs(title = "Pres
```



- Based on the first two graphs we can say that the **proportion of boys** is always **greater compared to girls in both the Arbuthnot and present data set.**
- While comparing induvidually we can observe that the **proportion of boys being born in Arbuthnot data set is greater/higher compared to present Data set**
- There is a **decline in the boy_ratio in the present data set.**

Exercise 7

In what year did we see the most total number of births in the U.S.?

```
# Insert code for Exercise 7 here
present %>%
arrange(desc(total))
```

```
## # A tibble: 63 x 6
##     year    boys   girls   total present_boy_ratio present_girl_ratio
##    <dbl>   <dbl>   <dbl>   <dbl>             <dbl>              <dbl>
##  1  1961 2186274 2082052 4268326             0.512              0.488
##  2  1960 2179708 2078142 4257850             0.512              0.488
##  3  1957 2179960 2074824 4254784             0.512              0.488
##  4  1959 2173638 2071158 4244796             0.512              0.488
##  5  1958 2152546 2051266 4203812             0.512              0.488
##  6  1962 2132466 2034896 4167362             0.512              0.488
##  7  1956 2133588 2029502 4163090             0.513              0.487
```

```
## 8  1990 2129495 2028717 4158212            0.512              0.488
## 9  1991 2101518 2009389 4110907            0.511              0.489
## 10 1963 2101632 1996388 4098020            0.513              0.487
## # i 53 more rows
```

In the year 1961 we saw the most total number of births in the U.S.

reference:

- for plotting two line plots in a single graph i learnt and picked up the code from (This website)[https://forum.posit.co/t/plotting-multiple-line/37056]